Jonathan Abraham, M.D., Ph.D.

Professor of Microbiology
Harvard Medical School

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Education and Employment:

2026 - Present	Professor	Microbiology	Harvard Medical School
2024 - Present	Investigator		Howard Hughes Medical Institute
2022 - 2026	Associate Professor	Microbiology	Harvard Medical School
2017 - 2022	Assistant Professor	Microbiology	Harvard Medical School
2015 - 2018	Clinical & Research Fellow	Infectious Diseases	Brigham & Women's Hospital / Massachusetts General Hospital
2013 - 2015	Intern & Resident	Medicine	Brigham & Women's Hospital
2006 - 2012	M.D., Ph.D.	HST/Biophysics	Harvard-MIT M.D.-Ph.D. Program
2001 - 2005	B.A.	Biochemical Sciences *(cum laude)*	Harvard University

Appointments at Hospitals/Affiliated Institutions:

2019 - Present	Associate Physician	Infectious Diseases	Brigham & Women's Hospital
2018 - Present	Associate Member	IDMP	Broad Institute
2018 - Present	Affiliate Faculty	Immunology	Harvard Medical School

Research Experience:

Harvard Medical School, Department of Microbiology

My lab uses methods in molecular virology, immunology, and structural biology to study how viruses interact with receptors on host cells and are neutralized by antibodies. We also study how viral polymerases replicate viral genomes and/or transcribe viral genes once cells are infected and how antivirals can target these proteins. We hope that the information we unearth through our basic, mechanistic studies of proteins that play critical roles in the life cycles of viruses could be harnessed to develop effective drugs and vaccines.

Selected manuscripts from our research:

Structure and operating principles of a monkeypox virus replisome.
Yu Z, Sathyanarayana P, Tan JMJ, Hu S, Fan X, Gao A, Kranzusch PJ, Loparo JJ, **Abraham J**.
2026. PMID: 42686903.

Mechanisms of HSV-1 helicase-primase inhibition and replication fork complex assembly.
Yu Z[#], Sathyanarayana P[#], Liu C, Tan JMJ, Yang P, Das B, Hu S, Fan X, Ji C, Weller SK, Shekhar M, Coen DM, Kranzusch PJ, Loparo JJ[*], **Abraham J**[*].
2026. PMID: 41468884.
[#] *Denotes co-first authors.*
[*] *Denotes co-corresponding authors.*

Molecular basis for shifted receptor recognition by an encephalitic arbovirus.
Fan X[#], Li W[#], Oros J, Plante JA, Mitchell BM, Plung JS, Basu H, Nagappan-Chettiar S, Boeckers JM, Tjang LV, Mann CJ, Brusic V, Buck TK, Varnum H, Yang P, Malcolm LM, Choi SY, de Souza WM, Chiu IM, Umemori H, Weaver SC, Plante KS, **Abraham J**.
2025. PMID: 40187345.
[#] *Denotes co-first authors.*

Honors and Prizes:

2024	Howard Hughes Medical Institute Investigator
2024	John and Virginia Kaneb Fellowship Award
2022	Charles E. W. Grinnell Fund for Medical Research
2022	Smith Family Foundation Odyssey Award
2022	Burroughs Wellcome Fund Investigators in the Pathogenesis of Infectious Disease (PATH) Award
2021	Star-Friedman Challenge for Promising Scientific Research
2021	Vallee Foundation Scholar Award
2017	Massachusetts Infectious Diseases Society Maxwell Finland Award
2016	NIH Director's Early Independence Award (EIA)
2016	Burroughs Wellcome Fund Career Award for Medical Scientists (CAMS)
2014	Burroughs Wellcome Fund Postdoctoral Enrichment Program Award
2008	American Society for Clinical Pathology Award for Academic Excellence and Achievement

Editorial Activities:

Ad hoc Reviewer

Nature, Science, Cell, Cell Reports, PNAS, New England Journal of Medicine, Science Translational Medicine, Nature Communications, PLoS Pathogens, Journal of Virology, Virology

Other Editorial Roles

eLife Board of Reviewing Editors (2020)

Committee Service:

Local

2020 - Present	Steering Committee	Massachusetts Consortium on Pathogen Readiness
2020 - Present	Executive Committee, Ph.D. Program in Virology	Harvard University
2018 - Present	Admissions Committee, Ph.D. Program in Virology	Harvard University
2022	Immunology Department Faculty Search Committee	Harvard Medical School
2017 - 2019	Dean's Faculty Council	Harvard Medical School
2018	Microbiology Department Faculty Search Committee	Harvard Medical School

Grant Review Activities:

2025	Personal Research Grants	Israel Science Foundation
2025	Expert Reviewer	UK Research Institute
2022	NIH Virology A study section ad hoc member	National Institutes of Health
2020	Personal Research Grants	Israel Science Foundation
2020	Gilliam Fellowship Stage-1 External Review	Howard Hughes Medical Institute

Teaching and Advising:

2018 - 2024	Co-course Director Virology 200 (Introduction to Virology)	Half-year course
2017 - Present	Qualifying Exam (PQE) Committees	9 students
2017 - Present	Thesis Committees	3 students
2009 - 2019	Leverett House Resident Tutor in Medicine	In-residence advisor at Harvard College, responsible for overseeing 30-40 students per semester
2017	Meet the HST-PI	One session

Clinical Practice Activities:

2018 - Present	Infectious Diseases Attending Brigham & Women's Hospital	4–6 weeks per year

Research Publications:

1. Yu Z, Sathyanarayana P, Tan JMJ, Hu S, Fan X, Gao A, Kranzusch PJ, Loparo JJ, **Abraham J**. Structure and operating principles of a monkeypox virus replisome. 2026. PMID: 42686903.

2. Yu Z[#], Sathyanarayana P[#], Liu C, Tan JMJ, Yang P, Das B, Hu S, Fan X, Ji C, Weller SK, Shekhar M, Coen DM, Kranzusch PJ, Loparo JJ[*], **Abraham J**[*]. Mechanisms of HSV-1 helicase-primase inhibition and replication fork complex assembly. 2026. PMID: 41468884.
   [#] *Denotes co-first authors.*
   [*] *Denotes co-corresponding authors.*

3. Fan X[#], Li W[#], Oros J, Plante JA, Mitchell BM, Plung JS, Basu H, Nagappan-Chettiar S, Boeckers JM, Tjang LV, Mann CJ, Brusic V, Buck TK, Varnum H, Yang P, Malcolm LM, Choi SY, de Souza WM, Chiu IM, Umemori H, Weaver SC, Plante KS, **Abraham J**. Molecular basis for shifted receptor recognition by an encephalitic arbovirus. 2025. PMID: 40187345.
   [#] *Denotes co-first authors.*

4. Mann CJ, Yang P, Olal D, Fan X, Smith KN, Clark LE, Krammer F, Bian Y[*], **Abraham J**[*]. Molecular organization of the New World arenavirus spike glycoprotein complex. 2025. PMID: 40781447.
   [*] *Denotes co-corresponding authors.*

5. Hu S[#], Kim H[#], Yang P[#], Yu Z, Ludeke B, Mobilia S, Pan J, Stratton M, Bian Y, Fearns R[*], **Abraham J**[*].
   Structural and functional analysis of the Nipah virus polymerase complex.
   2025. PMID: 39837328.
   [#] *Denotes co-first authors.*
   [*] *Denotes co-corresponding authors.*

6. Shankar S[#], Pan J[#], Yang P[#], Bian Y[#], Oroszlán G, Yu Z, Mukherjee P, Filman DJ, Hogle JM, Shekhar M, Coen DM, **Abraham J**.
   Viral DNA polymerase structures reveal mechanisms of antiviral drug resistance.
   2024. PMID: 39197451.
   [#] *Denotes co-first authors.*

7. Li W, Plante JA, Lin C, Basu H, Plung JS, Fan X, Boeckers JM, Oros J, Buck TK, Anekal PV, Hanson WA, Varnum H, Wells A, Mann CJ, Tjang LV, Yang P, Reyna RA, Mitchell BM, Shinde DP, Walker JL, Choi SY, Brusic V, Llopis PM, Weaver SC, Umemori H, Chiu IM, Plante KS, **Abraham J**.
   Shifts in receptors during submergence of an encephalitic arbovirus.
   2024. PMID: 39048821.

8. Yang P[#], Li W[#], Fan X, Pan J, Mann CJ, Varnum H, Clark LE, Clark SA, Coscia A, Basu H, Smith KN, Brusic V, **Abraham J**.
   Structural basis for VLDLR recognition by eastern equine encephalitis virus.
   2024. PMID: 39095394.
   [#] *Denotes co-first authors.*

9. Hale M, Netland J, Chen Y, Thouvenel CD, Smith KN, Rich LM, Vanderwall ER, Miranda MC, Eggenberger J, Hao L, Watson MJ, Mundorff CC, Rodda LB, King NP, Guttman M, Gale M, **Abraham J**, Debley JS, Pepper M, Rawlings DJ.
   IgM antibodies derived from memory B cells are potent cross-variant neutralizers of SARS-CoV-2.
   2022. PMID: 35938988.

10. Junqueira C, Crespo Â, Ranjbar S, de Lacerda LB, Lewandrowski M, Ingber J, Parry B, Ravid S, Clark S, Schrimpf MR, Ho F, Beakes C, Margolin J, Russell N, Kays K, Boucau J, Das Adhikari U, Vora SM, Leger V, Gehrke L, Henderson LA, Janssen E, Kwon D, Sander C, **Abraham J**, Goldberg MB, Wu H, Mehta G, Bell S, Goldfeld AE, Filbin MR, Lieberman J.
    FcγR-mediated SARS-CoV-2 infection of monocytes activates inflammation.
    2022. PMID: 35385861.

11. Tolan NV, Sherman AC, Zhou G, Nabel KG, Desjardins M, Melanson S, Kanjilal S, Moheed S, Kupelian J, Kaufman RM, Ryan ET, LaRocque RC, Branda JA, Dighe AS, **Abraham J**, Baden LR, Charles RC, Turbett SE.
    The effect of vaccine type and SARS-CoV-2 lineage on commercial SARS-CoV-2 serologic and pseudotype neutralization assays in mRNA vaccine recipients.
    2022. PMID: 35311584.

12. Xiong X[#], Tian S[#], Yang P[#], Lebreton F, Bao H, Sheng K, Yin L, Chen P, Zhang J, Qi W, Ruan J, Wu H, Chen H, Breault DT, Wu H, Earl AM, Gilmore MS[*], **Abraham J**[*], Dong M[*].
    Emerging enterococcus pore-forming toxins with MHC/HLA-I as receptors.
    2022. PMID: 35259335.
    [#] *Denotes co-first authors.*
    [*] *Denotes co-corresponding authors.*

13. Hickerson BT, Daniels-Wells TR, Payes C, Clark LE, Candelaria PV, Bailey KW, Sefing EJ, Zink S, Ziegenbein J, **Abraham J**, Helguera G, Penichet ML, Gowen BB.
    Host receptor-targeted therapeutic approach to counter pathogenic New World mammarenavirus infections.
    2022. PMID: 35091550.

14. Clark LE[#], Clark SA[#], Lin C[#], Liu J[#], Coscia A, Nabel KG, Yang P, Neel DV, Lee H, Brusic V, Stryapunina I, Plante KS, Ahmed AA, Catteruccia F, Young-Pearse TL, Chiu IM, Llopis PM, Weaver SC, **Abraham J**.
VLDLR and ApoER2 are receptors for multiple alphaviruses.
2022. PMID: 34929721.
[#] *Denotes co-first authors.*

15. Nabel KG[#], Clark SA[#], Shankar S[#], Pan J, Clark LE, Yang P, Coscia A, McKay LGA, Varnum HH, Brusic V, Tolan NV, Zhou G, Desjardins M, Turbett SE, Kanjilal S, Sherman AC, Dighe A, LaRocque RC, Ryan ET, Tylek C, Cohen-Solal JF, Darcy AT, Tavella D, Clabbers A, Fan Y, Griffiths A, Correia IR, Seagal J, Baden LR, Charles RC, **Abraham J**.
Structural basis for continued antibody evasion by the SARS-CoV-2 receptor binding domain.
2022. PMID: 34855508.
[#] *Denotes co-first authors.*

16. Ferrero S, Flores MD, Short C, Vazquez CA, Clark LE, Ziegenbein J, Zink S, Fuentes D, Payes C, Batto MV, Collazo M, García CC, **Abraham J**, Cordo SM, Rodriguez JA, Helguera G.
Antibody-based inhibition of pathogenic New World hemorrhagic fever mammarenaviruses by steric occlusion of the human transferrin receptor 1 apical domain.
2021. PMID: 34132574.

17. Wellner A, McMahon C, Gilman MSA, Clements JR, Clark S, Nguyen KM, Ho MH, Hu VJ, Shin JE, Feldman J, Hauser BM, Caradonna TM, Wingler LM, Schmidt AG, Marks DS, **Abraham J**, Kruse AC, Liu CC.
Rapid generation of potent antibodies by autonomous hypermutation in yeast.
2021. PMID: 34168368.

18. Sahtoe DD, Coscia A, Mustafaoglu N, Miller LM, Olal D, Vulovic I, Yu TY, Goreshnik I, Lin YR, Clark L, Busch F, Stewart L, Wysocki VH, Ingber DE, **Abraham J**[*], Baker D[*].
Transferrin receptor targeting by de novo sheet extension.
2021. PMID: 33879614.
[*] *Denotes co-corresponding authors.*

19. Clark SA[#], Clark LE[#], Pan J[#], Coscia A, McKay LGA, Shankar S, Johnson RI, Brusic V, Choudhary MC, Regan J, Li JZ, Griffiths A, **Abraham J**.
SARS-CoV-2 evolution in an immunocompromised host reveals shared neutralization escape mechanisms.
2021. PMID: 33831372.
[#] *Denotes co-first authors.*

20. Koma T, Huang C, Coscia A, Hallam S, Manning JT, Maruyama J, Walker AG, Miller M, Smith JN, Patterson M, **Abraham J**, Paessler S.
Glycoprotein N-linked glycans play a critical role in arenavirus pathogenicity.
2021. PMID: 33647064.

21. Clark LE, Mahmutovic S, Raymond DD, Dilanyan T, Koma T, Manning JT, Shankar S, Levis SC, Briggiler AM, Enria DA, Wucherpfennig KW, Paessler S, **Abraham J**.
Vaccine-elicited receptor-binding site antibodies neutralize two New World hemorrhagic fever arenaviruses.
2018. PMID: 29760382.

22. Gruszczyk J, Kanjee U, Chan LJ, Menant S, Malleret B, Lim NTY, Schmidt CQ, Mok YF, Lin KM, Pearson RD, Rangel G, Smith BJ, Call MJ, Weekes MP, Griffin MDW, Murphy JM, **Abraham J**, Sriprawat K, Menezes MJ, Ferreira MU, Russell B, Renia L, Duraisingh MT, Tham WH.
Transferrin receptor 1 is a reticulocyte-specific receptor for Plasmodium vivax.
2018. PMID: 29302006.

23. Mahmutovic S, Clark L, Levis SC, Briggiler AM, Enria DA, Harrison SC, **Abraham J**.
Molecular basis for antibody-mediated neutralization of New World hemorrhagic fever mammarenaviruses.
2015. PMID: 26651946.

24. Demogines A, **Abraham J**, Choe H, Farzan M, Sawyer SL.
Dual host-virus arms races shape an essential housekeeping protein.
2013. PMID: 23723737.

25. **Abraham J**, Corbett KD, Farzan M, Choe H, Harrison SC.
Structural basis for receptor recognition by New World hemorrhagic fever arenaviruses.
2010. PMID: 20208545.

26. Hood CL, **Abraham J**, Boyington JC, Leung K, Kwong PD, Nabel GJ.
Biochemical and structural characterization of cathepsin L-processed Ebola virus glycoprotein: implications for viral entry and immunogenicity.
2010. PMID: 20053739.

27. **Abraham J**[#], Kwong JA[#], Albariño CG, Lu JG, Radoshitzky SR, Salazar-Bravo J, Farzan M, Spiropoulou CF, Choe H.
Host-species transferrin receptor 1 orthologs are cellular receptors for nonpathogenic New World clade B arenaviruses.
2009. PMID: 19343214.
[#] *Denotes co-first authors.*

28. Radoshitzky SR[#], **Abraham J**[#], Spiropoulou CF, Kuhn JH, Nguyen D, Li W, Nagel J, Schmidt PJ, Nunberg JH, Andrews NC, Farzan M, Choe H.
Transferrin receptor 1 is a cellular receptor for New World haemorrhagic fever arenaviruses.
2007. PMID: 17287727.
[#] *Denotes co-first authors.*

29. Betts MR, Nason MC, West SM, De Rosa SC, Migueles SA, **Abraham J**, Lederman MM, Benito JM, Goepfert PA, Connors M, Roederer M, Koup RA.
HIV nonprogressors preferentially maintain highly functional HIV-specific CD8+ T cells.
2006. PMID: 16467198.

Reviews and Editorials:

1. **Abraham J**.
Monoclonal antibodies with extended half-life to prevent COVID-19.
2022. PMID: 35675183.

2. Coen DM, Lawler JL, **Abraham J**.
Herpesvirus DNA polymerase: Structures, functions, and mechanisms.
2021. PMID: 34861935.

3. **Abraham J**.
Passive antibody therapy in COVID-19.
2020. PMID: 32533109.

4. Choe H, Jemielity S, **Abraham J**, Radoshitzky SR, Farzan M.
Transferrin receptor 1 in the zoonosis and pathogenesis of New World hemorrhagic fever arenaviruses.
2011. PMID: 21807555.

Other Non-Peer-Reviewed Scholarship:

1. Rubin EJ, Baden LR, **Abraham J**, Morrissey S.
   Audio Interview: Viral evolution and the future of monoclonal antibodies.
   2022. PMID: 35704488.

2. Rubin EJ, Baden LR, **Abraham J**, Morrissey S.
   Audio Interview: The implications of changes in the structural biology of SARS-CoV-2.
   2021. PMID: 33704945.

Invited Research Presentations:

Local

2022 **Department of Microbiology Retreat**
Identification of conserved cellular receptors for divergent alphaviruses
Harvard Medical School
Falmouth, MA

2022 **Infectious Diseases and Microbiome Program**
Structural basis for antibody evasion by the SARS-CoV-2 spike protein
Broad Institute, Cambridge, MA

2021 **Virology Ph.D. Program Seminar Series**
Structural basis for continued antibody evasion by the SARS-CoV-2 receptor-binding domain
Harvard Medical School, Boston, MA

2021 **Program in Immunology Seminar Series**
Spike protein intra-host evolution reveals shared mechanisms of antibody neutralization escape
Harvard Medical School, Boston, MA

2021 **Harvard-MIT MD-PhD Program Annual Retreat**
Eva J. Neer Memorial Lecture
The shifting antigenic landscape of the SARS-CoV-2 spike protein
Cape Cod, MA

2020 **Biophysics Graduate Program Retreat**
Structural basis for potent SARS-CoV-2 antibody neutralization
Harvard University, Boston, MA

2020 **Structural Biology Retreat**
Structural basis for affinity maturation of a potent SARS-CoV-2 neutralizing antibody response
Harvard University, Boston, MA

2019 **MCB Cellular Dynamics Research Talk**
Receptor-binding and antibody neutralization of hemorrhagic fever arenaviruses
Harvard University, Cambridge, MA

2017 **Virology Seminar Series**
Neutralizing antibodies and viral hemorrhagic fevers
Harvard Medical School, Boston, MA

2017 **Infectious Disease & Microbiome Program**
Harnessing passive immunity to combat emerging viral threats
Broad Institute, Cambridge, MA

2017 **Center for Virology and Vaccine Research**
Molecular mechanism for antibody-mediated cross-neutralization of divergent hemorrhagic fever arenaviruses
Beth Israel Deaconess Medical Center, Boston, MA

2016 **Virology Graduate Program Annual Retreat**
Antibody neutralization of New World hemorrhagic fever arenaviruses
Harvard University
Beverly, MA

Regional

2021 **University of Massachusetts Chan Medical School**
Department of Microbiology and Physiological Systems
Shared mechanisms of SARS-CoV-2 antibody neutralization escape
Worcester, MA

2020 **Cell, Molecular & Developmental Biology Graduate Program**
Tufts University School of Medicine
Receptor binding and antibody neutralization of New World hemorrhagic fever arenaviruses
Boston, MA

2017 **Protein Engineering Summit (PEGS) Boston: Biologics and Vaccines for Infectious Diseases**
Antibody-based therapies against New World hemorrhagic fevers
Boston, MA

National

2023 **Department of Chemistry & Biochemistry Seminar Series**
Mechanisms of immune evasion by the SARS-CoV-2 spike protein
UCLA, Los Angeles, CA

2023 **Department of Microbiology Seminar Series**
Mechanisms of antibody evasion by the SARS-CoV-2 spike protein
University of Washington, Seattle, WA

2022 **iMicro Seminar Series**
Immune evasion by the SARS-CoV-2 spike protein
UCSF, San Francisco, CA

2022 **Department of Microbiology and Immunology Seminar Series**
SARS-CoV-2 spike protein evolution of immune escape
Cornell University College of Veterinary Medicine, Ithaca, NY

2022 **Vaccine Research Center Tuesday Seminar Series**
Convergent cellular receptors for divergent alphaviruses
NIH NIAID Vaccine Research Center
Bethesda, MD *(Virtual)*

2022 **Annual Harold Neu Symposium, The Infectious Diseases Society of New York**
Antibody evasion by the SARS-CoV-2 spike protein
Memorial Sloan Kettering
New York, NY *(Virtual)*

2021 **Department of Microbiology Seminar Series**
Shared mechanisms of SARS-CoV-2 antibody neutralization escape
Icahn School of Medicine at Mount Sinai
New York, NY *(Virtual)*

2021 **American Society for Virology Annual Meeting**
Convergent pathways for SARS-CoV-2 antibody neutralization escape
(Virtual)

2021 **Federation of Clinical Immunology Societies (FOCIS) Annual Meeting**
Shared mechanisms of SARS-CoV-2 antibody neutralization escape
(Virtual)

International

2026 **International Herpesvirus Workshop**
Mechanisms of HSV-1 helicase-primase inhibition and replication fork complex assembly
Montreal, Canada

2026 **Keystone Meeting—Viral Pathogenesis and Immunity**
Alphavirus interactions with cellular receptors
Hannover, Germany

2024 **Keystone Meeting—Positive Strand RNA Viruses**
How equine encephalitic alphaviruses interact with cellular receptors
Killarney, Ireland

2024 **49th Lorne Conference on Protein Structure and Function**
How alphaviruses sneak into cells
Lorne, VIC, Australia

2023 **Physical Virology Gordon Research Conference**
Mechanisms of immune evasion by the SARS-CoV-2 spike protein
Lucca (Barga), LU, Italy

2018 **Ninth International Virus Assembly Symposium**
Receptor-binding and antibody neutralization of New World hemorrhagic fever viruses
Plenary Presentation
Madeira, Portugal

2017 **Department of Structural Virology**
Cross-neutralization of New World hemorrhagic fever arenaviruses by receptor-binding site antibodies
Institut Pasteur, Paris, France

2017 **3rd International Conference on Vaccine Research & Development**
Vaccine-elicited broadly neutralizing antibodies against New World hemorrhagic fever arenaviruses
Reston, VA, United States